Activity 1

a) "the cat sat"

Trigram "the cat sat" appears **1 time**, and "the cat" (context) appears 2 times.
P(sat | the cat) = 1/2 = 0.5

Score = 0.5, Level = trigram (no backoff needed since the trigram exists)

b) "dog ate fish"

- Trigram "dog ate fish": doesn't exist → back off
- Bigram "ate fish": count = 1, context "ate" = 1 → P = 1/1 = 1.0, apply penalty since we backed off from trigram
- Score = 0.4 × 1.0 = 0.4

Score = 0.4, Level = bigram (backed off once from trigram)

c) "dog sat fish"

- Trigram "dog sat fish": doesn't exist → back off
- Bigram "sat fish": doesn't exist → back off again
- Unigram "fish": count = 1, total tokens = 19 → P = 1/19 ≈ 0.0526
- Apply penalty twice (backed off twice): Score = 0.4 × 0.4 × 0.0526 ≈ 0.0084

Score ≈ 0.0084, Level = unigram (backed off twice)

Activity 2

Part A: P(sat | the cat) with λ_1=0.1, λ_2=0.3, λ_3=0.6

- $P_{unigram}$ (sat) = 2/19 ≈ 0.1053
- P_{bigram} (sat | cat) = count(cat sat)/count(cat) = 1/2 = 0.5
- $P_{trigram}$ (sat | the cat) = count(the cat sat)/count(the cat) = 1/2 = 0.5

$P_{interp} = \lambda_1 \times 0.1053 + \lambda_2 \times 0.5 + \lambda_3 \times 0.5$
$= 0.1 \times 0.1053 + 0.3 \times 0.5 + 0.6 \times 0.5$
$= 0.01053 + 0.15 + 0.3$
$=$ **0.46053**

P(sat | the cat) ≈ 0.4605, using the formula P = λ1·P(unigram) + λ2·P(bigram) + λ3·P(trigram)

Part B: Compare λ sets for "the dog ate" and "dog ate fish"

Trigram 1: "the dog ate" (context "the dog")

- Unigram P(ate) = 1/19 ≈ 0.05263
- Bigram P(ate|dog): count(dog ate)=0, count(dog)=1 → 0/1 = 0
- Trigram P(ate|the dog): count(the dog ate)=0, count(the dog)=1 → 0/1 = 0

Trigram 2: "dog ate fish" (context "dog ate")

- Unigram P(fish) = 1/19 ≈ 0.05263
- Bigram P(fish|ate): count(ate fish)=1, count(ate)=1 → 1/1 = 1.0
- Trigram P(fish|dog ate): count(dog ate fish)=0, count(dog ate)=0 → 0 (undefined, treat as 0)

Set 1: λ = (0.1, 0.3, 0.6)

Trigram 1: P = 0.1×0.05263 + 0.3×0 + 0.6×0 = 0.005263
Trigram 2: P = 0.1×0.05263 + 0.3×1.0 + 0.6×0 = 0.305263

ln(0.005263) ≈ −5.2470
ln(0.305263) ≈ −1.1864

Sum of logs (Set 1) ≈ −6.4334

Set 2: λ = (0.3, 0.6, 0.1)

Trigram 1: P = 0.3×0.05263 + 0.6×0 + 0.1×0 = 0.015789
Trigram 2: P = 0.3×0.05263 + 0.6×1.0 + 0.1×0 = 0.615789

ln(0.015789) ≈ −4.1486
ln(0.615789) ≈ −0.4849

Sum of logs (Set 2) ≈ −4.6335

Activity 3: Entropy and Perplexity

Part A: Weather: Sunny=0.5, Cloudy=0.25, Rainy=0.25

$H = -\Sigma\, p \log_2 p$
$= -(0.5 \log_2 0.5 + 0.25 \log_2 0.25 + 0.25 \log_2 0.25)$
$= -(0.5\times(-1) + 0.25\times(-2) + 0.25\times(-2))$
$= -(-0.5 - 0.5 - 0.5)$
$=$ **1.5 bits**

Perplexity $= 2^H = 2^{1.5} \approx$ **2.828**

Entropy = 1.5 bits, Perplexity ≈ 2.828

Part B: Words after "the"

From the corpus, "the" is followed by: cat, mat, dog, rug, cat (again) → let's check carefully:
"the cat" (×2), "the mat" (×1), "the dog" (×1), "the rug" (×1)

Total count of "the" as context = 5 (matches unigram count of "the")

- P($^{cat}/_{the}$) = 2/5 = 0.4
- P($^{mat}/_{the}$) = 1/5 = 0.2
- P($^{dog}/_{the}$) = 1/5 = 0.2
- P($^{rug}/_{the}$) = 1/5 = 0.2

$H = -\Sigma\, p \log_2 p$
$= -(0.4 \log_2 0.4 + 0.2 \log_2 0.2 + 0.2 \log_2 0.2 + 0.2 \log_2 0.2)$

$\log_2 0.4 \approx -1.3219$
$\log_2 0.2 \approx -2.3219$

$H = -(0.4 \times (-1.3219) + 3 \times (0.2 \times (-2.3219)))$
$= -(-0.52877 - 1.39314)$
= **1.9219 bits**

Perplexity = $2^{1.9219} \approx$ **3.7925**

Words after "the": cat (p=0.4), mat (p=0.2), dog (p=0.2), rug (p=0.2). Entropy ≈ 1.922 bits, Perplexity ≈ 3.793

Entropy of next word after "sat":

"sat" is followed by: "on" (×2) — that's the only continuation in the corpus.
P(on|sat) = 2/2 = 1.0

$H = -(1.0 \times \log_2 1.0) = -(1.0 \times 0) =$ **0 bits**

Entropy after "sat" = 0 bits, because "on" is the only word that ever follows "sat" — there's no uncertainty at all (probability = 1.0).